Can Critical Periods in Language Models Be Reopened?

Abstract

Critical periods (windows after which a skill becomes hard to acquire regardless of further data) are usually studied in models whose pretraining history cannot be controlled. We train transformers of 11M–60M parameters from random initialization on a ~10M-word corpus, injecting synthetic arithmetic and long-distance *wh*-licensing on delay schedules that hold total skill exposure fixed and vary only *when* that exposure begins.

Delaying arithmetic exposure to the midpoint of training produces a persistent endpoint deficit against seed-matched early-exposure anchors: 0.214 $\pm$ 0.055 at 23M and 0.229 $\pm$ 0.037 at 60M, in the same direction at every seed. Long-distance *wh*-licensing shows no detectable deficit under five distinct probes, inverting our pre-registered prediction that syntax would show the *stronger* critical period. We claim this dissociation in absolute terms and not as differential susceptibility, because the two skills are not equally observable: the syntax anchor sits 0.028 above its metric's chance floor against arithmetic's 0.791, so a proportionally matched deficit would have fallen below the design's resolution (MDE 0.048). Four of the five probes ask whether a critical period exists *given the acquisition that occurred*; only a representational probe, which carries independent dynamic range (selectivity 0.60, $p < 1e\text{-}8$) and is flat across timing ($p = 0.48$): asks whether enough was acquired for one to have been detectable at all. That probe, not the count of nulls, is what the dissociation rests on.

Where the deficit exists, it reopens. Shifting the learning-rate schedule at the delay onset restores endpoint accuracy to 0.767 $\pm$ 0.014 at 23M and 0.796 $\pm$ 0.022 at 60M, against untreated nulls of 0.577 $\pm$ 0.051 and 0.600 $\pm$ 0.040: at 23M with no measurable cost to held-out perplexity, at 60M with a cost bounded above by 1.67 ppl that is plausibly an annealing-horizon artifact. Both recovered endpoints fall inside a pre-registered band, so recovery converges on a scale-invariant level rather than tracking capacity, and it collapses the cross-seed variance the untreated deficit shows. Concentrated re-exposure to the missed skill does not reproduce this: a data-only cram phase stays near the null, while cram combined with the schedule shift is indistinguishable from the shift alone. Closure at this onset is therefore largely a property of the optimizer's decayed learning-rate schedule rather than of entrenched representations.

The two results are timing effects at different levels: *which* skills a late start damages is skill-specific, while the damage itself is carried by the optimizer's schedule rather than by representations that have hardened against revision.

1 §1 Introduction

A critical period is a window during which experience shapes a system, after which the same experience has much less effect. Whether anything like it exists in language models is usually asked of models whose training history cannot be inspected: a released checkpoint has already seen whatever the web contained, in whatever order, so the timing of any particular skill's arrival is unknown and unmanipulable. Fine-tuning such a model tells us what it can still learn, not what it could have learned had the material arrived earlier.

We take the other route. Every model here is trained from random initialization on a fixed ~10M-word corpus, with a target skill injected into the training stream on a schedule we control. Conditions differ in *when* the skill arrives and in nothing else (total skill tokens are held constant), so a difference in final performance is a timing effect, not a dose effect. This costs us scale, the largest model being 60.3M parameters, and buys the one thing a pretrained checkpoint cannot provide: knowing when the model first saw arithmetic.

Within that design we find, first, that **delaying arithmetic exposure to the midpoint of train-**

ing leaves a deficit that the remaining half of training does not close, at both confirmatory sizes and at every seed. Second, that **long-distance *wh*-licensing shows no such deficit** under five distinct probes, inverting our pre-registered prediction that syntax, the more structurally scaffolded skill, would be the more fragile one. We state that dissociation in absolute terms rather than as differential susceptibility; §3.3 gives the measurement reason. Third, that **the deficit is reopenable**: restoring the learning-rate schedule at the point of delayed exposure recovers the endpoint to within 90 % of an early-exposure anchor at both sizes, and to nearly the same absolute level at each. Fourth, that **concentrated re-exposure does not reproduce this**: extra skill density on top of a restored schedule adds nothing, while a restored schedule without it recovers the skill. What closes the window at this onset is therefore largely the optimizer's decayed learning-rate schedule rather than representations hardened against revision.

Two things follow for how we report. A delayed model's endpoint is strongly seed-dependent while its early-exposure anchor is not, so we report absolute endpoints beside their nulls rather than recovery as a percentage of a gap whose denominator moves. And because several results here are nulls, we read non-significance off the minimum detectable effect rather than the p-value: two structurally identical size comparisons in this paper carry opposite evidential weight for that reason (§3.1, §5.3).

We registered three predictions before running these experiments: that syntax would show the stronger critical period, that embedding reset would be the most effective intervention, and that a reset-driven recovery would build new circuitry rather than repair the original. §7.2 states what happened to each. A reader holding the pre-registration will find all three answered there, including the one that could not be tested.

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2 §2 Methods

2.1 2.1 Models and corpus

All models are decoder-only transformers trained from random initialization; **no pretrained checkpoint is used anywhere in this work, including for baselines.** This is the design constraint that makes the rest of the paper meaningful: a model that has already passed through an uncontrolled pretraining history cannot be asked when it first met a skill.

tag	layers	heads	width	total params	non-embedding params
11M	4	8	256	11.54M	3.15M
23M	6	8	384	23.20M	10.62M
60M	8	10	640	60.30M	39.33M

The base corpus is the BabyLM "strict-small" collection (~10M words), tokenized with a byte-pair encoder trained on that corpus, vocabulary 16,384. Three configurations are used:

Both columns are given because untied embeddings make them tell different stories. At vocabulary 16,384 the input embedding and output projection are 4.19M parameters each at the smallest size (**72.7 % of that model, against 34.8 % at the largest**), so the three configurations span 5.2× in total parameters but 12.5× in the parameters that do computation. Size comparisons in this paper are stated in total parameters, which is the axis our pre-registered analyses used; readers translating our scale results into non-embedding terms should note that the effective range is wider than the labels suggest, and that the smallest model is unusually embedding-dominated (§2.5).

Context length is 512 tokens. Positional information is rotary (θ = 10,000); normalization is pre-norm; activations are GELU. **Input embeddings and the output projection are untied**: a deliberate choice, since one of our interventions resets the embedding layer and tied weights would confound an input-representation reset with a change to the prediction head. Attention is `F.scaled_dot_product_attention`, which dispatches to FlashAttention-2 kernels on our GPU. Training uses bf16 autocast, micro-batches of 8 × 512 tokens with 4 gradient accumulation steps, AdamW (β = 0.9/0.95, weight decay 0.1, gradient clipping 1.0), a peak learning rate of 6e-4 with 300 warmup steps and cosine decay to 0.1 × peak. `torch.compile` is available in the trainer but was **disabled for the runs reported here**. Every run reported here is 5,000 optimizer steps over 81.92M tokens. Checkpoint spacing varies by run family (100 steps for the 23M arithmetic runs, 1,250 at 11M and 60M, 2,500 for the size sweep); it sets how finely a trajectory can be re-read afterwards and nothing else, since delay and intervention triggers fire on exact step count.

2.2 2.2 Delay schedule and exposure matching

A run's data stream is a mixture of the base corpus and a skill stream. The skill stream has an **onset step** and a **density**, both fixed before training starts, and set so that total skill tokens are constant across delay conditions: a delay-0 run spreads its skill exposure across all 5,000 steps, while a delay-50 run delivers the same token count at higher density from step 2,500 onward. Onsets are specified as fractions of total steps and converted to exact step numbers before the run begins, so triggers fire on step count and never on wall-clock time.

Each run is identified by a `run_id` encoding skill, delay, exposure fraction, size and seed; checkpoints store optimizer state and a metadata dictionary alongside weights. **A `run_id` is the identity of a trained model, not a label for a configuration**, where two runs share a configuration but were trained separately, we treat them as different models and never average or interchange their numbers.

2.3 2.3 Measures

Three measurements recur, and each is fixed once here:

- **Arithmetic**: teacher-forced exact match at step 5,000 on 500 held-out problems.
- **Syntax**: a continuation-scored licensing interaction on *wh*-extraction minimal pairs.
- **General capability**: held-out validation perplexity, always reported as a three-number decomposition (early anchor, delayed null, treated arm) rather than as a single delta.

The syntax metric requires a word of justification, because the obvious one fails. Raw BLiMP-style *wh*-vs-*that* pairs differ by a single word, so their score is dominated by the unigram frequency prior for "that": an untrained-for-syntax model scores 0.000 on one paradigm and 1.000 on its complement while knowing nothing about gap licensing. Taking the interaction of the with-gap and no-gap conditions cancels that main effect (a purely frequency-driven model lands at exactly 0.500), so only genuine gap-licensing knowledge moves the score. All syntax numbers in this paper are that interaction, and 0.500 is its floor.

2.4 2.4 Eval-set hygiene

Exactly one arithmetic problem set is held out of training. An earlier version of our circuit analysis generated its evaluation items per-run, and those draws overlapped the training problems by 17–19 % in four of six cases; all affected results were re-scored on the held-out set, and the correction changed a conclusion (§5.2). On identical items our two evaluation harnesses agree to ~0.005, so accuracies from different harnesses are comparable to about 0.01 and no finer. We report this rather than only the corrected numbers, because the released logs contain both generations.

2.5 2.5 The smallest size is excluded by a pre-registered floor

Our pre-registration required an early-exposure anchor to reach 0.60 before a delay comparison at that size could be interpreted. The 11M anchor reaches 0.538 and has not plateaued, so its delay comparison is between two under-trained models and is **excluded by the pre-registered floor as a capacity/budget boundary**. Its numbers are reported in §3.1 because they are nominally significant and a reader recomputing them will find them; they do not support a claim in either direction about a critical period at that scale.

2.6 2.6 Analysis designs that were planned and not run

We pre-registered three analyses: a mixed-effects model of final accuracy on delay × size, a factorial model of recovery percentage on intervention × skill × delay, and a model of circuit overlap on intervention × skill. None is estimable on the design that survived. After the floor exclusion the first has twelve observations with one per seed × cell and no within-cell replication for a random effect. The second has a banned denominator (recovery percentage divides by the seed-variable null), and two of its three factors are constants. The third has both factors constant. We replaced them with per-seed paired tests, an absolute-endpoint gate, and a rank correlation with exact permutation p, respectively, and say so here rather than presenting the replacements as the original plan.

2.7 2.7 Multiplicity

Confirmatory tests are Holm-corrected within a declared family and reported both uncorrected and adjusted: the Detect family (paired deficit at two sizes), the Reopen family (lift over the matched null at two sizes), and the agreement-slope family (the paired attraction−complexity contrast, one test per mixture). Gates whose thresholds were fixed in advance are decision rules, not tests, and carry

size	seed	early (d0)	delayed (d50)	deficit
23M	1	0.798	0.524	0.274
	2	0.792	0.626	0.166
	3	0.784	0.582	0.202
	mean ±	**0.791 ±**	**0.577 ±**	**0.214 ±**
	SD	**0.007**	**0.051**	**0.055**
60M	1	0.798	0.574	0.224
	2	0.840	0.646	0.194
	3	0.848	0.580	0.268
	mean ±	**0.829 ±**	**0.600 ±**	**0.229 ±**
	SD	**0.027**	**0.040**	**0.037**

no p. Exploratory analyses are flagged in-text and never carry a headline.

Membership is declared positively, so that a test's absence from a family is a statement rather than an omission. §6 is the section where this matters most: it reports many p values on models trained for a different purpose, and exactly one of its families is confirmatory. Its header says so, and names the rest as exploratory by declaration rather than leaving the reader to infer scope from which numbers happen to carry an adjustment.

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3 §3 Detect

3.1 3.1 Delayed arithmetic exposure produces a persistent endpoint deficit

Every model in this section sees the same number of arithmetic tokens; the conditions differ only in *when* those tokens enter the stream. The delayed condition (d50) begins arithmetic exposure at the midpoint of training, the early condition (d0) at step 0, and both are read at step 5000 on the same 500 held-out problems. Because seeds are crossed (the same seed gives the same initialization in both conditions) the design-appropriate statistic is the per-seed paired deficit, tested separately at each size and Holm-corrected across the two confirmatory sizes.

Table 1. Arithmetic exact-match accuracy at step 5000, per seed and pooled.

The deficit is present at every seed at both sizes: paired $t(2) = 6.74$, $p = 0.0107$ at 23M and $t(2) = 10.64$, $p = 0.0044$ at 60M, both surviving Holm correction within the Detect family (adjusted 0.0107 and 0.0087; Hedges $g = 2.22$ and 3.51). The pre-registered Welch gate, which required a gap of at least 0.15, is cleared at both sizes (60M one-sided $p = 0.0010$). A model that receives its arithmetic exposure late does not catch up by the end of training, despite having received all of it.

The deficit is of **similar magnitude at both sizes**: the 60M–23M difference is 0.015, Welch $p = 0.72$, with a minimum detectable effect at 80 % power of **0.144**. That null constrains (it excludes any size effect larger than about two-thirds of the deficit itself), but it is not a claim that size is irrelevant, and it should not be read as one. We return to this in §5, where a structurally identical size comparison carries the opposite evidential weight for one reason only: its MDE exceeds the entire range its outcome occupies. The two nulls are best read as a pair.

The severity of the deficit is seed-variable, and the variability lives on the delayed side. The early anchor is tight across seeds (SD 0.007 at 23M) while the delayed endpoint is not (SD 0.051), so per-seed deficits range from 0.166 to 0.274 at 23M against a baseline that barely moves. Seed 1 is the deepest-hit on both readings (lowest delayed endpoint and largest deficit), and reproduces as such when retrained from scratch. This is why every recovery result in §4 is reported as an absolute endpoint beside its null rather than as a percentage of the gap: the gap is a seed-variable denominator, so a recovery-percentage is unstable by construction even when the underlying effect is not.

The 11M models are excluded by a pre-registered floor. Their paired deficit is nominally significant (0.181 ± 0.087, $p = 0.0345$, $g = 1.19$), and we report it because a reader recomputing our numbers will find it. It is excluded because the 11M early anchor (0.538 ± 0.018) never plateaued and sits below the pre-registered 0.60 floor: the comparison is between two under-trained models, so the difference cannot be attributed to timing rather than to capacity or budget. This is a capacity boundary, not evidence that no critical period exists at 11M.

Intermediate onsets (d25, d55, d75) were run at 23M seed 1 only. They are plotted for completeness without a fitted trend; at $n = 1$ per onset they describe this run and support no claim about the shape of the timing curve.

3.2 3.2 Long-distance *wh*-licensing shows no detectable critical period

The same design applied to long-distance *wh*-licensing yields nothing. The five probes below are **five different hypotheses about where an effect could hide, not five tests of one hypothesis**; each carries its own positive control or power statement, and the argument is that pairing rather than the count.



Figure 1: Placeholder for the §3 Detect figure.

1. **Endpoint by delay** ($n = 3$): no gap, against a delay-0 baseline that itself ranges 0.520–0.586 over eight seeds (a spread exceeding any candidate effect) on the same harness that resolves arithmetic's 0.214.
2. **Acquisition dynamics**: Welch $p = 0.46$, underpowered alone and carried by probe 3.
3. **Rate / bimodality**: a pre-registered compound gate over **72 runs** (24 seeds $\times$ 3 sizes) returns DISCONFIRM: the mean rises monotonically with size ($0.528 \rightarrow 0.538 \rightarrow 0.552$) but fails its floor at threshold 0.56 (top size $0.208 < 0.40$) and its trend at 0.57 ($p = 0.074$), passing only at the permissive 0.55, and this powered test, sized in advance by simulation, is why long-distance syntax left the main grid.
4. **Representational probe**: delay-0 versus delay-50 $p = 0.48$ *while the same probe detects the structure at selectivity 0.60, p < 1e-8*: the positive control passes.
5. **Learning-rate budget** (hilr50, $n = 3$): ΔLD = $+0.011 \pm 0.005$, with Δlocal exceeding ΔLD at every seed and validation perplexity improving: a non-specific whole-model lift, not a rescue of the delayed skill.

Probe 4 is the load-bearing one, for a reason that the other four cannot supply. Probes 1, 2, 3 and 5 all ask whether a critical period is visible *in the behaviour that was acquired*. Probe 4 asks whether the underlying structure is encoded at all, and finds that it is, and that its encoding does not depend on when exposure began. **The structure is learned; what is timing-invariant is the readout.**

This null is scoped to $\leq$ 60.3M parameters, the ~10M-word BabyLM budget, and the continuation-scored licensing-interaction metric. Outside that scope it says nothing.

	arithmetic	syntax LD
early-exposure anchor	0.791 ± 0.007	0.528 (24 seeds)
chance floor	~0 (exact match)	0.500 (by metric construction)
headroom above floor	**0.791**	**0.0283**
observed deficit	0.214 (27.1 % of headroom)	none detected
a proportionally matched deficit would be	—	0.0077 (0.49 seed-SD)
MDE at 80 % power, n = 3/group	—	**0.0482**

3.3 3.3 Skill dissociation

Under one design, one pipeline and matched total exposure, delayed exposure costs arithmetic 0.214 ± 0.055 and costs long-distance *wh*-licensing nothing measurable across five probes. That contrast is the paper's central detection-side result, and it inverts our pre-registered prediction that syntax would show the *stronger* critical period: a prediction we restate with its outcome as a standalone item in §7.2, since a reader holding our pre-registration will look for it there rather than here.

It must be stated in absolute terms, and not as differential susceptibility, because the two skills are not equally observable. The asymmetry is large enough to state as a table rather than to note as a caveat:

Table 2. Why the dissociation is scoped.

The long-distance design's MDE **exceeds its entire headroom by 1.7**$\times$. A total collapse of the skill to chance would be a change of 0.0283 (smaller than the smallest effect the design could resolve), and a susceptibility proportionally identi-

cal to arithmetic's would require $n \approx 67$ per group. The behavioural null therefore cannot, on its own, separate *syntax is less susceptible* from *syntax had no room to show susceptibility*.

Two things follow, and only one of them is a rescue. The learning-rate-budget null (probe 5) does **not** relieve the problem: it establishes that the ceiling is not an optimization-budget artifact, which makes the ceiling real and immovable by that lever, and so *preserves* the dynamic-range limit rather than lifting it. The representational probe (probe 4) does: it is the one readout not bounded by the 0.0283 headroom, it carries genuine dynamic range, and it is flat across timing. Had delayed exposure damaged the underlying representation, that probe had the range to register it and did not.

The differential-susceptibility reading has now been tested on the rate readout, and it returns a bound rather than a decision. Under the mean-endpoint readout it remains unmeasurable at the n achieved here. We therefore ran the rate-collapse test at 60M with **24 delayed seeds** against the existing 24-seed baseline — 24 rather than the 14 a full-abolition assumption prices, because at 14 the test clears α only on an exact 0/14, and one high seed anywhere lands it at $p = 0.18$. The two arms are superimposed: 0.5528 ± 0.0138 delayed against 0.5518 ± 0.0131 at baseline, with the high mode at 6/24 in **both**, Fisher exact $p = 0.630$ and threshold-free Mann–Whitney $p = 0.626$ (both Holm-adjusted to 1.000), and a mean difference of +0.0010 [−0.0064, +0.0087].

The result is asymmetric by construction and is reported that way: it **excludes a full collapse** of the high mode (power 1.000) and **does not exclude a half collapse** (power 0.188; roughly 59 seeds per group would be needed). At 23M the same test needs 82 seeds and remains out of reach (§8). It does not become a further route to the syntax null: under our cross-method rule (§10) a probe that agrees but is underpowered on the contrast at issue is not corroboration, and this one is underpowered against exactly the alternative that matters. It bounds the dissociation's scope rather than promoting it.

Finally, a note on what this result is not. Our pre-registration anticipated a *reopening*-side dissociation: some skill/intervention combinations recover and others do not. What the data support is a *detection*-side one: syntax never showed a deficit, so no syntax reopening was attempted. The two are superficially the same finding and are not the same claim, and we report the one we have.

size	treated (`lr_shift`)	delayed (null)	early anchor	gate (0.90 × own d0)	margin	lift, p
23M	**0.767 ± 0.014**	0.577 ± 0.051	0.791 ± 0.007	0.712	+0.055	0.0065
60M	**0.796 ± 0.022**	0.600 ± 0.040	0.829 ± 0.027	0.746	+0.050	0.0027

4 §4 Reopen

4.1 4.1 An LR-schedule shift restores the delayed endpoint

All intervention arms fork from a single saved checkpoint at step 2500 (identical weights, identical optimizer state, identical step), so that arms differ only in the intervention applied. The `lr_shift` arm restores the learning-rate schedule that a delay-0 model would have been on at the point delayed exposure begins, then anneals normally. Success was pre-registered as a two-part condition: the treated endpoint must reach 90 % of that run's *own* early-exposure anchor, and general capability must not degrade beyond a matched threshold.

Table 3. Endpoints at step 5000. The null is shown beside the treated arm throughout; neither is collapsed into a ratio.

Per-seed treated endpoints are 0.752 / 0.778 / 0.772 at 23M and 0.796 / 0.818 / 0.774 at 60M. Every seed at both sizes clears its gate, and the per-seed lift over the matched null survives Holm correction within the Reopen family (adjusted 0.0065 and 0.0054).

Two features of this table are more surprising than the recovery itself.

The recovered endpoint is scale-invariant. Both treated means fall inside the pre-registered same-endpoint band [0.722, 0.812], despite early anchors that differ by 0.038 and a 2.6× difference in parameters. Recovery converges on a level rather than tracking capacity.

The intervention collapses cross-seed variance. The null's SD (0.051 at 23M, 0.040 at 60M) drops to the treated arm's (0.014, 0.022). The seed-variable severity documented in §3.1 is a property of the *untreated* delayed models; the intervention makes the outcome both higher and more reproducible.



Figure 2: Placeholder for the §4 Reopen figure.

4.2 4.2 The recovery does not come at the cost of general capability

Perplexity is reported as the three-number decomposition (early anchor a, delayed null b, treated arm c), because a single delta cannot distinguish a treated arm that improved from a null that degraded.

At **23M there is no breach**: $c - b = -0.889$ / -0.204 / $+0.193$ across seeds, against a pre-registered gate of +1.0 (seed 1: $b = 50.747$, $c = 49.858$, both from the canonical `...seed1_recheck` lineage). Two of three seeds improve.

At **60M the gate is tripped in all three seeds**: $c - b = +1.670 \pm 0.104$ ppl. We report it as a trip and honour it as written. Two controls bound what it can mean. A no-op fork reproduces the null on both perplexity and accuracy (Δppl -0.012, Δacc +0.011), so fork mechanics contribute nothing. And at **matched learning rate** the treated arm *leads* the null by 2.478 ± 0.240 ppl (a lead larger than the breach), because the arm is read mid-anneal at roughly six times the null's terminal learning rate, with about 4.15 ppl of annealing gain still unrealized. The breach persists only if the arm's remaining anneal delivers less than 1.67 ppl of that. The evidence leans toward near-full erasure, but the anneal slope is unmeasured, so we quote **no point estimate**: the residual cost is bounded above by 1.67 ppl and a partly real cost is not excluded. Extending the arm past step 5000 is the run that would collapse this bound in either direction (§8).

4.3 4.3 The effect is specific to the delayed skill

The same intervention applied to delayed *syntax* models (replicated at all three seeds, matching the depth of the arithmetic arm) moves nothing: Δ local = -0.024 / -0.004 / $+0.044$ and Δ LD = -0.021 / $+0.007$ / $+0.015$, sign-inconsistent around zero at every seed. A single-seed control could not have distinguished "no generic effect" from "an effect that happened to land near zero once", which is why the control was run at the arm's own seed depth. Whatever `lr_shift` does, it is not a generic late-training lift.

4.4 4.4 Closure at this onset is largely a property of the schedule

The remaining Tier-1 interventions have been extended to $n = 3$ at 23M (seeds 1–3), forked from the same common checkpoints. Each is scored per seed against **its own fork source**, whose cross-seed endpoint is 0.571 ± 0.061 with range [0.506, 0.626]. That is a different lineage from the Detect table's null and is flagged here rather than reconciled, per the §5 rule-2 convention: same run_id, same model; different run_id, different model.

Only freeze supports a directional statement, and it now does so at three seeds rather than one. For the other four the reading keeps its label and changes its basis. At $n = 1$ they were undecidable because a single seed cannot be separated from a null whose own spread is 0.10 wide; at $n = 3$ they are undecidable because their per-seed lifts **disagree in sign**. A near-zero mean with disagreeing signs is an absence of effect, not a small effect obscured by pooling — the same rule we apply to cram below. They remain *not* evidence that those interventions fail.

The extension earned its cost in two places. Seed 1 gave the largest lift of any seed for **every one of the four**, and two endpoints moved materially once

arm (23M, n = 3)	endpoint	per-seed lift	verdict
embedding reset	0.585 ± 0.001	+0.080 / −0.040 / +0.002	no consistent direction
reset + LR spike	0.571 ± 0.042	+0.070 / −0.100 / +0.028	no consistent direction
LR spike alone	0.578 ± 0.048	+0.030 / +0.004 / −0.014	no consistent direction
cram (concentrated re-exposure)	0.577 ± 0.005	+0.068 / −0.052 / 0.000	no consistent direction
freeze	**0.422 ± 0.081**	−0.174 / −0.182 / −0.092	**harmful**: below the entire null range at every seed

the other seeds were in: LR spike from 0.536 to 0.578 ± 0.048, and freeze from 0.332 to 0.422 ± 0.081. Freeze's verdict is unchanged, but its single-seed figure was an extreme rather than a typical draw. Embedding reset was our pre-registered favourite, and §7.2 states that prediction and its outcome as a standalone item.

The one arm we extended to $n = 3$ decides the mechanism. Concentrated re-exposure alone leaves the endpoint at 0.574: at the null. Combined with the schedule shift it reaches 0.781 ± 0.031, far above cram alone and indistinguishable from the shift by itself: the per-seed differences are −0.006 / +0.016 / +0.032, straddling zero with no consistent sign. (We report the signs rather than the +0.014 mean, because a near-zero mean with disagreeing signs is an absence of effect, not a small effect obscured by pooling.)

So additional skill density on top of a restored schedule adds nothing, while a restored schedule without additional density recovers the skill. **At this onset, what closes the window is largely the optimizer's decayed learning-rate schedule rather than representations that have hardened against revision.** The scope of that statement is d50, arithmetic, and these two sizes; it is not a claim that entrenchment plays no role at other onsets.

A final arm belongs in the record without supporting a claim. The 11M `lr_shift` arm passes its gate (0.747 against a gate of 0.484) but the pass is degenerate: the gate is 90 % of an anchor that failed the §3.1 floor, and the arm *exceeds its own early anchor* (0.538), which indicates that the 11M anchor was learning-rate-starved rather than that

schedule	early (d0)	delayed (d50)	paired deficit
cosine	0.791 ± 0.007	0.577 ± 0.051	0.214 ± 0.055
WSD	0.851 ± 0.016	0.858 ± 0.022	−0.007 ± 0.013

reopening over-recovers. The 11M arm enters the paper only through its mechanism decomposition, where the delayed lift of +0.389 splits into a generic component (+0.115) and an exposure-alignment component (+0.274): about 70 % of the effect requiring the delayed exposure profile. That decomposition is exploratory and carries no headline.

4.4.1 4.5 The mechanism replicates on a second schedule

Every run above uses warmup with cosine decay, so §4.4's claim rests on a single schedule shape. Warmup-stable-decay (WSD) separates the readings, because the two schedules disagree exactly where the experiment intervenes: at the d50 onset (step 2500) cosine has decayed to 3.570e-4, 59.5 % of peak, while WSD is still at its 6.00e-4 peak and stays there until step 4000. Peak, floor and warmup are shared; only the shape between them differs. If closure is a decayed-LR phenomenon the deficit should largely vanish under WSD; if it persists near 0.21, §4.4 needs rewriting. Both outcomes were reportable, and the second is why the test was run before submission rather than left to a reviewer.

The deficit is abolished. Per-seed deficits under WSD are +0.002 / +0.000 / −0.022, straddling zero, which meets the pre-registered *abolished* criterion on both of its clauses (mean ≤ 0.05 **and** inconsistent signs) rather than the weaker *substantially reduced*. The WSD-versus-cosine difference is −0.221 (Welch $p = 0.016$, Holm 0.032) against a pre-registered MDE of 0.092, and is read off the MDE rather than off the p.

Two limits were fixed before the data. The *presence* of a WSD deficit is well resolved; the *equality* of the two deficits is not, so this is never written as "the deficit is the same under both schedules". And WSD's higher endpoints on both arms are descriptive only: the deficit is a within-schedule paired contrast, so a cross-schedule level difference is not evidence about it. The result licenses "replicates on a second, deliberately dissimilar schedule" and not "schedule-general" — one non-cosine schedule, one size, one onset, one skill.

4.4.2 4.6 Level reopens; shape pays for it

`lr_shift` moves the learning rate's magnitude and its trajectory together, so §4.4's result is a state-

cell	endpoint	lift	Δppl
C — flat at late level	0.563 ± 0.063	-0.009	+1.70
A — level only	0.752 ± 0.034	+0.181	+1.86
B — shape only	0.575 ± 0.046	+0.004	+1.13
`lr_shift` — both	0.767 ± 0.014	+0.196	-0.33

ment about the pair rather than about either. Crossing level against shape separates them. The fourth cell of that 2×2 *is* `lr_shift`, so three arms were run, each at $n = 3$ from the same common checkpoints, with level matched on the window mean.

Three contrasts fall out along the design's own axes. **A against C** holds the shape flat and moves the level: +0.181 against −0.009, so the level is what reopens. **B against C** holds the level late and moves the shape: +0.004 against −0.009, so shape alone recovers nothing. **A against `lr_shift`** holds the level early and moves the shape: the lifts are near-identical, but the perplexity costs are +1.86 and −0.33, so the shape is what makes the recovery free. C is load-bearing rather than decorative — without it, A's lift is confounded with the absence of decay.

This resolves why one arm reopened and the others did not. Under §2's operational definition, recovery bought at the cost of general capability is forgetting rather than reopening, and A breaches the +1.0 ppl threshold; so A is not a reopening, and `lr_shift` remains the only arm that is, because it is the only cell supplying both components. The Holm-adjusted p values (A 0.092, `lr_shift` 0.075) are not the basis of either verdict: the reopening gate is a pre-registered decision rule and carries no p.

5 §5 Did the intervention repair the original circuit, or build a new one?

Recovery tells us the skill came back; it does not tell us whether it came back *in the same machinery*. For each seed and size we compare a triad that shares an initialization: **A**, the early-exposure model; **B**, the delayed model; and **C**, the `lr_shift`-recovered model forked from B. Each model's arithmetic circuit is localized by mean-ablating attention heads one at a time and recording exact-match damage. The decisive test is reciprocal: ablating **A's** top heads inside **C** measures how much of A's circuit is still load-bearing in the recovered model, and ablating **C's own** top heads inside C controls for C simply being robust to any ablation. We report the transfer ratio (the damage A's heads do in C as a fraction of the damage they do in A) where 1.0 means C inherited A's circuitry intact and 0 means A's heads are causally inert in C. One of the two size comparisons below is the underpowered counterpart to §3.1's well-powered one; §5.3 makes that pairing explicit.

All numbers are computed on the single held-out problem set. An earlier version of this analysis drew its eval items per-run, which overlapped training by 17–19 % in four of the six triads; all six were re-scored, the seed-1 triads reproduce exactly as a no-op control, and the correction is reported in §2.4 because it changed a conclusion.

5.1 5.1 Only one of four probes discriminates on this contrast

Before reading anything off the ablation numbers, it matters which methods can tell the two outcomes apart *at all*. Four were tried and three cannot:

- **Whole-layer mean-ablation saturates**, ablating any core layer drives accuracy to approximately zero in every model, so it separates nothing. Localization is therefore done at head granularity.
- **Linear CKA** of the residual stream sits at 0.66–0.83 across all six triads regardless of what the causal test says. Because the triads share an initialization, global geometry stays similar even where the wiring differs; CKA would have scored the most reorganized triad in the set at 0.658 and missed it entirely.
- **Single-head activation patching** is flat: its restoration ratio spans 0.591–0.792 (SD 0.083) against the ablation transfer's 0.187–0.868 (SD 0.265), and the two are uncorrelated ($r = 0.045$, $p = 0.93$; recomputing on the corrected transfers gives $r = -0.032$, so the flatness is not an artifact of the contamination). The cause is a floor: single-head restoration is 0.05–0.15 in every model, so the ratio divides two small numbers and has no resolving power. An earlier draft cited patching as independent corroboration on the strength of one seed; running it on all six withdrew that.
- **Pruning-based localization** was prescreened and rejected: mask overlap spans only 0.088 across the triads, and its agreement with ablation *flips sign* across the sparsity sweep. A method whose sign is a hyperparameter choice cannot adjudicate (Appendix A).

size	seed	deficit	transfer	control	Jaccard(A, C)	pre-registered verdict
23M	2	0.182	0.659	1.50	0.00	mixed / partial
60M	2	0.202	0.404	2.43	0.00	mixed / partial
23M	3	0.210	0.868	1.16	0.20	mixed / partial
60M	1	0.224	0.445	2.18	0.09	mixed / partial
60M	3	0.260	0.201	5.13	0.00	new circuit
23M	1	0.292	0.187	5.36	0.00	new circuit

We therefore report this cluster as **single-method by demonstration**, not by assumption. Three non-discriminating probes is itself the evidence for that label: the alternative, citing agreement from a method never shown to resolve the contrast, is how the retracted patching claim happened.

5.2 5.2 Recovered circuits span a continuum, and it runs backwards at the shallow end

Table 4. Reciprocal ablation transfer, ordered by deficit depth. Transfer = damage from A's heads in C ÷ their damage in A. Control = damage from C's own heads ÷ damage from A's heads in C, so a high control confirms C has a working circuit of its own.

Recovered models do not behave alike. They span a continuum from near-complete transfer of the original circuit to full reorganization. At one end (23M seed 3) the recovered model inherits 87 % of the damage A's circuit takes, and its own top heads are barely more damaging than A's: the original machinery is still carrying the skill. At the other (23M seed 1, 60M seed 3) A's heads are close to inert in C while C's own heads are five times more damaging: the skill returned in different machinery. **Every 60M triad sits at or below 0.445**, so at the larger size reorganization is what we observe at every seed.

We report this axis continuously and assign no categorical verdict. Our implementation would have labelled a triad *repaired* only on a conjunction of high transfer and high top-head-set overlap; the overlap condition is never met here (the maximum is 0.20), so the label was unreachable. It was an implementation default rather than a registered criterion, and relabelling under a corrected rule would be choosing a decision rule after seeing the number it decides (Appendix B). The mismatch is itself the methodological point: a recovered model's head set can overlap its baseline's almost not at all while the baseline's heads remain causally load-bearing: near-zero overlap at transfer 0.868. **Set overlap and causal overlap are different measurements**, and only the second answers the question. That is the case for reciprocal ablation over any similarity-of-components measure, and the same failure that makes CKA useless here, from a second direction.

The ordering suggests that deficit depth grades this outcome (deeper deficits reorganize more), but the evidence does not survive its own robustness check. Across the six triads Spearman $\rho = -0.714$ with an exact permutation $p = 0.0681$, and leave-one-out is fragile: four of the six single-point deletions push p above 0.05. The direction persists and is not resolvable at $n = 6$. More triads at either size are the only thing that would settle it, and this is the one place in the project where additional runs would still move a headline.

One pair of triads shows the relationship running backwards, and it is the best-measured comparison in the table. At 23M, seeds 2 and 3 have nearly the same deficit (0.182 vs 0.210), yet the *deeper* one shows *more* repair (0.868 vs 0.659), with non-overlapping bootstrap intervals: [0.832, 0.903] against [0.611, 0.708]. Those intervals reflect eval-item variance with one model per point (they say the two measurements differ, not that two populations do), but within the shallow-deficit regime the severity ordering inverts. Whatever grades circuit identity is not deficit depth alone.

5.3 5.3 On size, this design is underpowered, unlike the one in §3.1

Transfer averages 0.571 ± 0.349 at 23M and 0.350 ± 0.131 at 60M (Welch $p = 0.39$). **That non-significance means nothing here.** The minimum detectable effect at 80 % power is 0.809, against an outcome span of only 0.681 across all six triads: no difference this design could have produced would have been detected. The question is *untested*, not answered, and it is no longer unidentifiable, since the corrected deficit ranges at the two sizes overlap, so size is separable from deficit depth in a way it was not before the contamination fix.

This is worth setting beside §3.1's size comparison, which is structurally identical (same $n = 3$ per group, same Welch test, same two sizes, also non-significant), and carries opposite weight. There, the MDE (0.144) sits below the deficit being asked about (0.21), so the null genuinely excludes a large size effect. Here it exceeds the entire measured range. The two together are the clearest illustration of a rule this paper applies throughout: non-significance is read off the minimum detectable effect, never off the p.

What the size axis does show is **variance**. The 60M transfers are tightly clustered (SD 0.131) where the 23M ones are not (SD 0.349), a 2.7× ratio, mirroring §4's finding that the intervention collapses cross-seed variance in the endpoint. We report this as suggestive: it is a single comparison at $n = 3$ per group on an observational rather than manipulated factor.

5.4 5.4 What these results do not speak to

Our third pre-registered prediction linked circuit identity to the reset intervention: if reset were the winning intervention, we expected it to build a new circuit. Reset never produced recovery (§4.4), so the antecedent never occurred and the prediction was never entered. The models diagnosed here are `lr_shift`-recovered, which that prediction did not cover; §7.2 states this rather than letting the section read as an answer to it.

Three further quantities appear in our released logs and support nothing. The within-60M Spearman is mechanical at $n = 3$. An extrapolation from the 23M triads onto the 60M points was built before the contamination correction and no longer fits (residuals -0.30, -0.15, -0.20). Numbers from the causal-tracing and pruning scripts were computed on the contaminated draws and were never re-run, since neither supports a surviving claim. Circuit redundancy and sufficiency are likewise untested here: the retracted patching analysis was the only probe that addressed them.

6 Related Work

Critical periods. The concept comes from development, where early sensory deprivation produces deficits that later normal input does not repair (Hubel and Wiesel, 1970; Lenneberg, 1967; Johnson and Newport, 1989). The closest machine-learning analogue is Achille et al.'s demonstration that deep networks have critical learning periods: a deficit imposed early in training (blurred images) leaves a permanent gap even given unlimited clean data afterwards, and the window's timing is a property of training rather than of architecture (Achille et al., 2019). That work manipulates *input quality* in vision models; we manipulate *when a skill's data arrives* in a language model while holding its total quantity fixed, which is what lets a deficit be attributed to timing rather than to dose or degraded signal.

Plasticity loss and warm starting. A separate literature finds that networks become harder to train as training proceeds: warm-started models generalize worse than freshly initialized ones given the same final data (Ash and Adams, 2020), and continual-learning agents progressively lose the ability to fit new tasks, recoverable by re-injecting plasticity through resets or noise (Dohare et al., 2024; Lyle et al., 2023; Berariu et al., 2021). **This is the closest prior explanation of our reopening result and the one we most risk rediscovering**, so we state the difference precisely: those settings involve a distribution shift, a restart, or a sequence of distinct tasks, and typically recover plasticity by perturbing *weights*. Our models train once, without interruption, on a stationary mixture whose only manipulated variable is a skill's onset step, and the intervention that recovers the skill changes no weights at all, only the learning-rate schedule, while resetting weights (the standard plasticity remedy) produced no detectable recovery in our setting at $n = 1$. The mechanisms may still be the same one seen from two directions; what we can say is that the recovery here is obtainable without weight perturbation. (That reset arm is $n = 1$; §7.2 states what it does and does not license.)

Curriculum, data ordering, and delayed generalization. Ordering effects have a long history (Bengio et al., 2009) with mixed results at LM scale, where reordering often yields little; our finding is compatible with that, since total exposure is what matters for most of the corpus and it is one late-arriving skill that pays a timing cost. Delayed generalization on arithmetic is also studied as grokking (Power et al., 2022), which is a different phenomenon despite the surface adjacency: grokking is a long plateau followed by sudden generalization at *fixed* data, whereas ours is a permanent endpoint gap produced by *shifting* when the data arrives.

Small-scale LM evaluation. We use the BabyLM setting (Warstadt et al., 2023) and BLiMP-

style minimal pairs (Warstadt et al., 2020), with the metric modification of §2.3, and the agreement-attraction literature (Linzen et al., 2016; Marvin and Linzen, 2018) for §6.

Circuit identity. Our diagnosis uses causal ablation and activation patching in the circuit-analysis tradition (Elhage et al., 2021; Wang et al., 2023; Meng et al., 2022) and representational similarity (Kornblith et al., 2019); §5.1 reports the negative methodological result that three of those four probes do not discriminate on our contrast.

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7 §6 A secondary probe: agreement attraction across scale (exploratory except where stated)

Multiplicity scope for this section, declared here rather than inferred. One family in §6 is confirmatory: **F6**, the paired attraction−complexity contrast, one test per mixture, Holm-corrected at $m = 2$ (the syntax mixture survives; the arithmetic one does not). Every other p reported below — the attraction magnitude at a single size, the three mismatch-versus-chance tests, the complexity-slope null, the per-size cross-mixture contrasts, and the within-run trajectory — is **exploratory by declaration**: uncorrected, descriptive, and carrying no headline. They are reported because withholding them would misrepresent what was looked at, not because any of them is being asserted. This section analyses models trained for a different purpose, so its tests were not part of the pre-registered design and are not promoted by consistency with it.

The models trained for this study support one question the critical-period design did not set out to ask. Subject–verb agreement is acquired almost perfectly here (0.98 on controlled items), yet fails when a number-mismatching noun intervenes. A fully-crossed item design, in which the verb-form frequency prior cancels *exactly*, separates two components of that failure: **attraction**, the cost of an intervening noun whose number mismatches the head, and **complexity**, the cost of any intervening material at all.

Attraction is real, large, and the dominant component. At 23.2M it costs +0.303 ± 0.064 (p = 8.6e-05) against complexity's +0.143 ± 0.035, in both number directions and both frame types. A unigram-frequency control on the same items returns exactly 0.000 by construction, so the effect cannot be a verb-form prior. Attraction exceeds complexity at every size and is positive in **81 of 81 models** across two independent training mixtures: 72 syntax-enriched (3 sizes × 24 seeds) and 9 arithmetic-enriched.

Whether these models are anti-syntactic is scale-dependent. Mismatch accuracy on controlled items is 0.405 at 11.5M (below chance, $p < 0.001$), 0.511 at 23.2M (at chance, $p = 0.435$), and 0.610 at 60.3M (above chance, $p < 0.001$). Below-chance agreement occurs only at the smallest size.

Attraction declines with scale; whether it dissociates from complexity depends on the mixture. In the syntax-enriched mixture attraction falls **−0.316 per decade of parameters** ($r = -0.803$, p = 2.3e-17, $n = 72$), from +0.430 to +0.201, while complexity is **flat** (+0.016/decade, $p = 0.43$). The second mixture reproduces the attraction half in direction and magnitude, **−0.207/decade**, stable at −0.21 to −0.24 under every estimator tried but at **estimator-dependent significance (p = 0.033 under OLS, 0.073 under a heteroscedasticity-robust SE**) on an unbalanced (3, 3, 8) seed design whose leverage sits on the under-sampled small end. It does **not** reproduce the dissociation: complexity falls there too, and the paired within-model contrast is flat (−0.095, $p = 0.36$) against −0.333 (p = 3.8e-11) in the syntax mixture, the one part of the comparison that is not estimator-fragile. A cross-mixture divergence at 60.3M **does not survive replication** at $n = 8$ (+0.065, $p = 0.091$, halved from +0.133) and **is not excluded**: the 95 % CI still contains the original effect. An English-exposure explanation for it is excluded separately: exposure is constant across sizes by design and can only produce a uniform offset.

Within a single run, attraction may also decline over training, but we do not claim it. Between steps 2500 and 5000 the isolated attraction cost falls +0.392 → +0.303 (paired, $n = 6$ at 23.2M, 5/6 seeds): marginal at $p = 0.053$, with 53 % power against this effect size and an MDE of 0.125; roughly ten seeds would decide it. The mismatch cell itself improves significantly over the same interval (+0.070, $p = 0.041$). We report the direction and the power, and treat the trajectory as suggestive only.

The standard metric gets this wrong at every scale, in three different ways. Uncontrolled BLiMP-style items report below-chance behaviour at all three sizes, which is true only at the smallest; the frequency-controlled metric locates it correctly;

and only the crossed design separates attraction from complexity. Three metrics, three conclusions, on the same models. A related trap is worth naming: pooling all 72 models into one attraction estimate averages across a size-dependent effect and yields a number that describes no size: an averaging-across-a-dissociation error, not a confound.

This is also why the skill sits here rather than in §3. Agreement is unusable as a delayed-exposure target: it saturates by step 2500 and its supporting structure appears in roughly 39 % of the corpus against *wh*-extraction's 1.5 %, so it cannot meaningfully be withheld. Being *learnable enough to measure* and being *delayable* are different properties, and a design like ours needs both.

8 §7 Discussion

8.1 7.1 Two timing results at different levels

This paper reports two findings about timing, and they operate at different levels. The first says that *which* skills a late start damages is skill-specific: arithmetic pays a persistent cost for a delayed start, long-distance *wh*-licensing pays nothing we can measure. The second says that where the damage does occur, it is carried by the optimizer's learning-rate schedule rather than by representations that have hardened against revision, restoring the schedule restores the skill, and adding skill data without restoring the schedule does not. A late start hurts some skills and not others; when it hurts, what has closed is the model's step size, not its mind.

8.2 7.2 The three pre-registered predictions

Prediction #1 (that syntax would show the *stronger* critical period) was inverted. Arithmetic carries the effect at both confirmatory sizes; long-distance *wh*-licensing shows no detectable deficit under five distinct probes. We had reasoned that long-distance dependencies, being scaffolded by earlier-acquired structure, would be the more fragile to a late start. The opposite skill turned out to be the fragile one, and this inversion is the origin of the dissociation in §3.3 rather than an incidental miss.

Prediction #2 (that embedding reset would be the most effective intervention) was not borne out. Reset produced no detectable recovery, landing at 0.586 against a null whose cross-seed range is [0.524, 0.626]; the learning-rate shift was the only arm to clear the pre-registered gate. Because reset was run at a single seed, this is a statement about what we could detect, not a demonstration that reset fails: one seed cannot be separated from a null that spans 0.10. The prediction's second half (that reset would carry the largest forgetting risk) was also not observed as stated.

Prediction #3 (that a reset-driven recovery would build new circuitry) was never testable, because its antecedent did not occur. The prediction was conditional on reset being the winning intervention. Reset never produced recovery, so the conditional was never entered. §5 diagnoses models recovered by the learning-rate shift, which prediction #3 did not cover; those results should not be read as confirming or disconfirming it. We state this explicitly because §5 occupies the position in the paper where a reader holding our pre-registration will look for #3's answer.

8.3 7.3 What the mechanism result does not say

Schedule-driven closure is demonstrated at one onset (the midpoint of training), for one skill, at two model sizes. It is not a claim that representational entrenchment plays no role in critical periods generally, nor that it plays no role at later onsets in this same setup. Our intermediate-onset runs exist at a single seed and are not evidence about the shape of the timing curve.

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9 §8 Limitations

Five specific experiments would move a label, and we name them with their costs rather than gesturing at future work.

The dissociation's stronger reading is now bounded, not un-run. Whether syntax is genuinely *less susceptible* to delayed exposure, rather than merely having too little headroom to show susceptibility, was decidable by a rate-collapse test at 60M, and §3 reports it: 24 delayed seeds against our 24-seed baseline, returning 6/24 in both arms. That excludes a full collapse of the high-acquisition mode and leaves a half collapse open, which would need roughly 59 seeds per group. The remaining half is now the cheapest experiment that could promote the dissociation past its current scope; the same test at 23M requires 82 seeds and stays out of reach.

The 60M perplexity cost is bounded, not estimated. Extending the treated arm past step 5,000 to complete its annealing would convert our upper

bound of 1.67 ppl into a point estimate in either direction.

The circuit-identity gradient is unresolved at $n = 6$. Additional triads at either size are the only thing that would settle whether deficit depth grades circuit identity; the direction persists but four of six leave-one-out deletions break its significance.

The circuit results are single-method by demonstration. Joint head-group patching, shown to have dynamic range on the repair-versus-reorganization contrast, would let those claims drop the annotation. Three probes we tried do not discriminate.

One agreement-attraction slope is estimator-dependent. Three or more further seeds at the smallest size would rebalance the leverage that makes it significant under OLS and marginal under a robust standard error.

Beyond these: one corpus, one tokenizer, one language, a maximum of 60.3M parameters, and a single delay onset carrying the reopening result. Five of six intervention arms are single seed. The smallest size is excluded by a floor rather than solved.

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10 §9 Conclusion

Timing of exposure matters in these models, but not uniformly and not for the reason we expected. Delayed arithmetic exposure leaves a deficit that further training does not close; delayed long-distance *wh*-licensing leaves nothing we can detect, which inverts our prediction and which we state in absolute terms because the two skills are not equally observable. Where the deficit exists it reopens, and it reopens when the learning-rate schedule is restored rather than when the missed data is concentrated, so what closes at this onset behaves like a property of the optimizer rather than of entrenched representation. Whether the skill contrast reflects genuine differential susceptibility remains open, and we have named the fourteen-seed experiment that would decide it.

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